

Aditi Roy Adri

Computer Science & Engineering — BRAC University, Dhaka

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

OBJECTIVE

Motivated CSE student with a strong foundation in algorithms, machine learning, systems programming, and full-stack development. Seeking opportunities to apply technical expertise, contribute to impactful projects, and grow in a collaborative environment.

EDUCATION

BRAC University, Dhaka

2022 — Present

Bachelor of Science in Computer Science and Engineering

CGPA: 3.84 Credits Completed: 120 / 136

Holy Cross College, Dhaka

2019 — 2021

Higher Secondary Certificate (HSC)

GPA: 5.00

Narayanganj High School and College, Narayanganj

2017 — 2019

Secondary School Certificate (SSC)

GPA: 5.00

TECHNICAL SKILLS

Programming Languages

Python, C++, C, Java, PHP, Assembly (Emu8086 / x86), Bash

Web Technologies

HTML5, CSS3, JavaScript, React.js

ML & Data Science

PyTorch, NumPy, Pandas, Scikit-Learn, Matplotlib, Jupyter Notebook

Databases

MySQL

Graphics & Networking

OpenGL (PyOpenGL), Cisco Packet Tracer

Tools & Platforms

Git, GitHub, LaTeX, Linux, VS Code

Soft Skills

Communication, Critical Thinking, Problem-Solving, Teamwork

PROJECTS

Pet Image Segmentation & Breed Classification (U-Net)

[GitHub](#) ↗

Python · PyTorch · Jupyter Notebook · ResNet-34 · Oxford-IIIT Pet Dataset

- Built U-Net and Attention U-Net architectures for binary pet image segmentation and 37-breed classification. Achieved ~88% IoU using a pretrained ResNet-34 encoder; from-scratch models reached ~80% IoU. Implemented data augmentation, combined segmentation + classification loss, and ablation studies.

Roaming Sonic — Travel & Tour Management Platform

[GitHub](#) ↗ [Live](#) ↗

MongoDB · Express.js · React.js · Node.js · Vercel

- Full-stack MERN web platform for Bangladesh tourism featuring tour packages, hotel bookings, bus ticketing, guide services, AI chatbot, and referral rewards system. Live-deployed on Vercel.

Enterprise Network Simulation

[GitHub](#) ↗

Cisco Packet Tracer · VLSM · RIPv2 · DHCP · DNS · SMTP/POP3

- Designed a multi-site hybrid enterprise network implementing VLSM subnetting, RIPv2 dynamic routing, static route redistribution, floating static routes for WAN failover, DHCP, DNS, and SMTP/POP3 email services.

Metadata Journal System

[GitHub](#) ↗

C · Operating Systems

- OS-level implementation of file metadata management and journaling mechanisms to maintain filesystem consistency across crashes and concurrent access.

RISC-V ISA Converter

[GitHub ↗](#)

Python · RISC-V Architecture

- Bidirectional converter between RISC-V 32-bit assembly instructions and binary/hexadecimal machine code representations, supporting all standard RISC-V instruction formats (R, I, S, B, U, J).

GapTime — Schedule Sync Tool

[GitHub ↗](#) [Live ↗](#)

React.js · JavaScript · GitHub Pages

- React web application for finding and synchronizing free time gaps across multiple schedules. Deployed live on GitHub Pages.

EcoGrow Nursery Management System

[GitHub ↗](#)

PHP · MySQL · HTML/CSS

- Web-based management system for nursery order processing, event scheduling, and administrative control panel with role-based access.

Heart Disease Prediction — ML

[GitHub ↗](#)

Python · Scikit-Learn · Pandas · Matplotlib

- Comparative study of Logistic Regression, SVM, Random Forest, and KNN classifiers for heart disease prediction using clinical patient data, with feature analysis and model evaluation.

Space Shooter & Golden Rush — 3D OpenGL Games

[Space Shooter ↗](#) [Golden Rush ↗](#)

Python · PyOpenGL · Real-Time Rendering

- Two 3D games featuring enemy AI, collision detection, health systems, multiple camera perspectives, and real-time shader-based rendering built from scratch using PyOpenGL.

Minibank & Micro-Bkash — Assembly Simulations

[Minibank ↗](#) [Micro-Bkash ↗](#)

x86 Assembly · Emu8086 · Microprocessor Systems

- Two low-level simulations in Emu8086 assembly: a banking system with account management and a digital payment transaction simulator modelled on bKash.

Comparator & Mixed-Signal Generator

[Comparator ↗](#) [Signal Gen ↗](#)

Digital Logic Design · Analog Electronics · Op-Amp Circuits

- Designed and simulated a digital comparator circuit (DLD); separately built an op-amp based mixed-signal generator capable of producing square, triangular, and sine waveforms.

Web Utilities — QR Generator & Speed-Time Calculator

[QR Demo ↗](#) [Speed Demo ↗](#)

HTML · JavaScript · GitHub Pages

- Two lightweight browser tools deployed on GitHub Pages: an instant QR code generator for URLs/text, and a playback speed-to-duration calculator for media content.

Codeforces Solutions (Python)

[GitHub ↗](#)

Python · Competitive Programming

- Ongoing collection of **50+ Codeforces solutions** (76+ commits) spanning beginner to intermediate difficulty, demonstrating consistent algorithmic problem-solving practice.

EXPERIENCE & ACTIVITIES

- Competitive programming on Codeforces — 50+ problems solved in Python across multiple difficulty levels.
- Participation in university-level and inter-university programming contests at BRAC University.

INTERESTS

Machine learning, image processing, computer graphics, competitive programming, full-stack web development, movies and TV series.

LOCATION

Present: DIT Project, Merul Badda, Dhaka-1212 · **Permanent:** 57/2, Kalirbajar, Narayanganj